

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	VAISHNAVI.S
Register Number	192521313

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I VAISHNAVI.S (192521313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student:

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1.A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Introduction

In cloud computing, businesses often experience sudden increases in website traffic during special events such as seasonal sales, festive offers, or product launches. If the application cannot handle the increased number of users, the website may become slow or unavailable, leading to customer dissatisfaction and revenue loss. Cloud computing provides technologies such as auto-scaling, load balancing, and monitoring tools to efficiently manage such traffic spikes. These technologies ensure high availability, better performance, and a seamless user experience.

Cloud Auto-Scaling

Cloud auto-scaling is the process of automatically increasing or decreasing the number of computing resources based on application demand. During seasonal sales, thousands or even millions of customers may visit an e-commerce website simultaneously. If a fixed number of servers is used, they may become overloaded, resulting in slow response times or server crashes. Auto-scaling continuously monitors resource utilization such as CPU usage, memory consumption, and network traffic. When the workload exceeds a predefined threshold, it automatically launches additional virtual machines or cloud instances to handle the increased demand. After the traffic decreases, the extra resources are automatically removed to reduce operational costs.

Advantages of Auto-Scaling

- Automatically adjusts resources according to demand.
- Prevents server overload.
- Maintains consistent application performance.
- Reduces infrastructure cost by removing unused resources.
- Ensures efficient utilization of cloud resources.

Load Balancing

Load balancing is the process of distributing incoming user requests across multiple servers so that no single server becomes overloaded. A load balancer acts as an intermediary between users and servers. Whenever customers access the website, the load balancer receives the requests and forwards them to the server with available capacity. If one server becomes busy or fails, the load balancer automatically redirects traffic to healthy servers. This ensures that all servers share the workload equally and continue operating efficiently even during heavy traffic.

Advantages of Load Balancing

- Prevents server overload.
- Improves application performance.
- Increases system availability.

To Auto-Scaling and Load Balancing Work Together

Auto-scaling and load balancing complement each other to provide a highly available cloud environment.

1. Customer traffic suddenly increases.
2. Monitoring tools detect the increase in resource usage.
3. Auto-scaling launches additional cloud servers.
4. The load balancer distributes incoming requests evenly across all available servers.
5. If traffic decreases after the sale, auto-scaling removes unnecessary servers.

Reducing Downtime

Downtime occurs when a website or application becomes unavailable to users. Auto-scaling and load balancing significantly reduce downtime by:

- Adding additional servers before existing servers become overloaded.
- Redirecting traffic away from failed or unhealthy servers.
- Eliminating single points of failure.
- Maintaining continuous availability of cloud services.
- Recovering automatically from unexpected traffic spikes.

Improving User Experience

User experience is one of the most important factors for online businesses. Cloud auto-scaling and load balancing improve user experience by:

- Providing faster page loading.
- Reducing waiting time.
- Preventing website crashes.
- Ensuring smooth product searches.
- Supporting uninterrupted online payments.
- Maintaining quick checkout processes.

Role of Monitoring Tools

- CPU utilization
- Memory usage
- Network bandwidth
- Disk usage
- Response time

Monitoring tools generate alerts whenever abnormal conditions are detected. They also trigger auto-scaling policies when predefined thresholds are reached. For example, if CPU utilization exceeds **80%**, the monitoring system automatically initiates the creation of additional cloud instances. Similarly, when CPU utilization decreases below a specified limit, unnecessary servers are removed.

Real-World E-Commerce Scenario

- Load balancers distribute customer requests evenly across all available servers.
- If any server fails, traffic is immediately redirected to healthy servers.
- After the sale ends, auto-scaling removes extra servers to reduce infrastructure costs.

Because of these cloud technologies, Amazon can continue serving millions of users with minimal downtime while maintaining fast website performance and secure transactions.

Benefits of Using Auto-Scaling, Load Balancing, and Monitoring

- Improves response time.
- Ensures high availability.
- Optimizes cloud resource utilization.
- Reduces operational costs.
- Enhances customer satisfaction.
- Supports business continuity.
- Increases overall application reliability.

Conclusion

Cloud auto-scaling, load balancing, and monitoring tools work together to provide an efficient, reliable, and highly available cloud infrastructure. Auto-scaling automatically adjusts computing resources according to demand, while load balancing distributes user requests evenly across multiple servers. Monitoring tools continuously track system performance and trigger scaling actions whenever required. In real-world e-commerce platforms such as Amazon, these technologies ensure uninterrupted service, fast response times, reduced downtime, and an excellent user experience during seasonal sales and other high-traffic events.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Introduction

Organizations generate large amounts of important data every day, including customer information, financial records, business documents, and application data. Losing this data due to hardware failure, cyberattacks, or natural disasters can cause serious financial and operational problems. Cloud storage provides a secure, reliable, and cost-effective solution for storing backup data. It offers high durability, data redundancy, strong security mechanisms, and disaster recovery capabilities while reducing storage costs compared to traditional storage systems.

Cloud Storage Solutions

Cloud storage is a service that allows organizations to store, manage, and retrieve data over the internet using cloud service providers. Instead of purchasing and maintaining physical storage devices, organizations can store their backup data in cloud data centers. Cloud storage services automatically manage storage capacity, backup, maintenance, and data availability, allowing organizations to focus on their business operations.

Benefits of Cloud Storage

- Scalable storage capacity.
- High availability.
- Easy access from anywhere.
- Automatic backup.
- Reduced maintenance effort.
- Cost-effective storage solution.

Ensuring Durability

Durability refers to the ability of cloud storage to preserve data without loss over a long period. Cloud providers ensure durability by:

- Storing multiple copies of data.
- Regularly checking data integrity.
- Automatically replacing corrupted or damaged data.
- Using reliable storage hardware.
- Performing continuous backup and maintenance.

Even if a storage device fails, another copy of the data remains available, preventing permanent data loss.

Advantages of Durability

- Protects important business information.
- Minimizes the risk of data loss.
- Ensures long-term data preservation.
- Provides reliable backup storage.

Data Redundancy

Data redundancy means storing multiple copies of the same data in different storage devices or different geographic locations. Cloud providers automatically replicate backup data across multiple servers and data centers. If one server or an entire data center becomes unavailable due to hardware failure or natural disasters, another copy can be used immediately.

Benefits of Data Redundancy

- Prevents data loss.
- Ensures continuous availability.
- Improves fault tolerance.
- Supports disaster recovery.
- Increases business continuity.

Encryption for Data Protection

Encryption converts readable data (plaintext) into an unreadable format (ciphertext) using encryption algorithms. Only authorized users with the correct decryption key can access the original information. Cloud providers generally provide:

Encryption at Rest

- Protects data stored in cloud storage.
- Prevents unauthorized access if storage devices are stolen.

Encryption in Transit

- Protects data while it is transferred between users and the cloud.
- Prevents attackers from intercepting sensitive information during transmission.

Benefits of Encryption

- Maintains confidentiality.
- Protects sensitive business information.
- Prevents unauthorized data access.
- Ensures secure cloud communication.

Cloud Storage Cost Advantages

- No need to purchase storage hardware.
- Pay only for the storage actually used.
- No maintenance cost.
- Automatic software updates.

Disaster Recovery Use Case

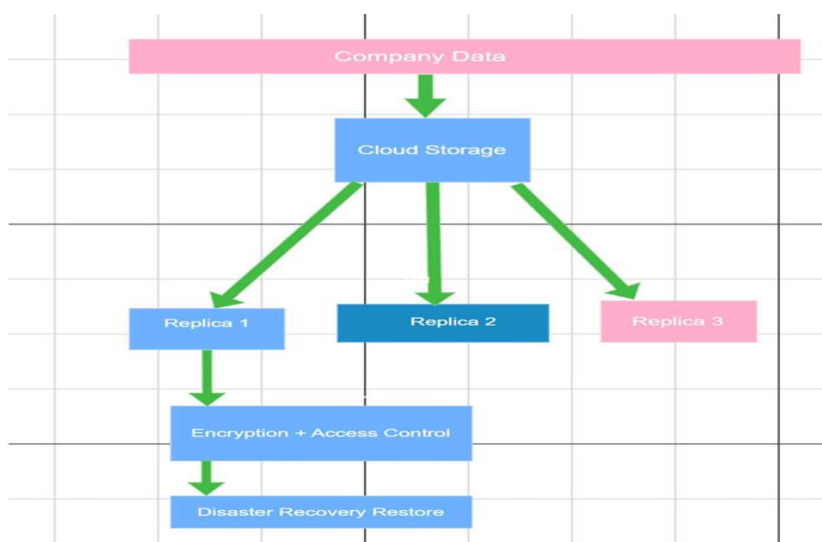
Consider a company that stores all its business data in cloud storage. One day, a flood damages the company's office and destroys its local computers and storage devices. This demonstrates how cloud storage supports effective disaster recovery and business continuity. Since backup copies are already stored securely in multiple cloud data centers:

Advantages of Cloud Storage for Backup

- High durability.
- Data redundancy.
- Secure encryption.
- Strong access control.
- Easy scalability.

Conclusion

Cloud storage is an efficient solution for securely storing large volumes of backup data. It ensures high durability by maintaining reliable copies of data and data redundancy by replicating information across multiple servers and locations. Encryption protects data during storage and transmission, while access control ensures that only authorized users can access sensitive information.



3.A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Introduction

When a mobile application is deployed globally, users from different countries may access the application from different locations. If the application data is stored in only one central server, users who are far away from that server may experience slow response times and higher latency. Cloud technologies such as Content Delivery Networks (CDNs), edge servers, and global cloud infrastructure help applications deliver faster and more reliable services to users worldwide.

Content Delivery Networks (CDNs)

A Content Delivery Network (CDN) is a distributed network of servers located in different geographical regions that store and deliver application content closer to users. Instead of sending every request to a single main server, CDNs deliver data from the nearest available server location. CDNs improve application performance by caching frequently accessed content such as videos, images, software files, and application resources on multiple servers around the world.

- Minimizing the physical distance between users and servers.
- Reducing network congestion.
- Providing faster content delivery.

Role of Edge Servers in Improving User Experience

Edge servers are servers placed closer to end users at different locations around the world. They act as intermediate points between users and the main cloud servers. Edge servers enhance user experience by processing and delivering data locally instead of depending completely on a central data center. This reduces the time required for data transfer and provides faster responses.

The benefits of edge servers include:

- **Lower latency:** Users receive responses faster because data travels a shorter distance.
- **Faster loading:** Application content loads quickly, improving user satisfaction.
- **Better reliability:** If one server location experiences issues, users can connect through another edge location.
- **Improved performance:** Reduces the workload on the main cloud servers by handling requests locally.

Cloud Providers Supporting Global Availability

Cloud providers support global availability by maintaining multiple data centers across different regions worldwide. These regions allow applications to run closer to users and provide continuous service.

- **Multiple availability zones:** Applications can run across different data centers to prevent failures.
- **Global load balancing:** User requests are automatically directed to the nearest and healthiest server.
- **Data replication:** Application data is copied across multiple regions for reliability.
- **Auto-scaling:** Resources are automatically increased during high traffic periods.
- **Disaster recovery:** Services can continue even if one region experiences an outage.

Real-Time Streaming/Gaming Scenario

Consider a global online gaming application such as a multiplayer mobile game. Millions of players from different countries connect to the game simultaneously. If all players connect to one central server, users located far away may experience delays, slow responses, and poor gameplay. Using CDNs and edge servers, game updates, graphics, and other resources are delivered from nearby servers. Cloud providers deploy gaming servers in multiple regions and use global load balancing to connect players to the nearest available server. Similarly, video streaming platforms use CDNs to deliver videos from nearby servers, allowing users to watch content without buffering.

For example:

- A player in India connects to an Asian edge server instead of a server located in the USA.
- Game data is delivered faster with reduced latency.
- Players experience smoother gameplay and faster response times.
- Multiple cloud regions ensure the game remains available even if one region fails.

Conclusion

CDNs, edge servers, and global cloud infrastructure play an important role in deploying mobile applications worldwide. CDNs improve performance by caching and delivering content from nearby locations, while edge servers reduce latency and provide faster responses. Cloud providers ensure global availability through multiple regions, data replication, load balancing, and disaster recovery mechanisms. In real-time streaming and gaming applications, these technologies provide smooth performance, low delay, and a better user experience for users across the globe.

4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Introduction

Financial institutions handle highly sensitive information such as customer details, account information, payment records, and transaction data. Due to the importance of this information, banks and financial organizations must follow strict security standards and government regulations while using cloud services. Cloud providers offer various security features such as regulatory compliance support, auditing mechanisms, logging, monitoring, encryption, and data location controls to ensure secure cloud usage.

Cloud Provider Support for Regulatory Compliance and Auditing

- **Compliance certifications:** Cloud providers maintain certifications such as ISO standards, SOC reports, and industry-specific security standards.
- **Security policies:** They provide tools to configure access control, identity management, and data protection.
- **Regular audits:** Cloud platforms conduct security audits to verify that systems follow required regulations.
- **Compliance reporting:** Organizations can generate reports to prove that security requirements are being followed.

Role of Logging in Cloud Security

Logging records all activities and events occurring within cloud systems. It provides a detailed history of user actions, system changes, and security events. Logs can record:

- User login activities.
- Data access attempts.
- File modifications.
- Configuration changes.
- Failed authentication attempts.
- Network activities.

The importance of logging includes:

- Helps detect unauthorized access.
- Supports investigation after security incidents.
- Provides evidence during audits.
- Helps identify suspicious activities.

For example, if an employee accesses confidential customer transaction records without authorization, logs can help identify who accessed the data and when it occurred.

Role of Monitoring in Cloud Security

Monitoring continuously observes cloud resources and application performance to identify potential threats and security issues.

Monitoring tools track:

- Unusual user behavior.
- Network traffic patterns.
- System performance.
- Security alerts.
- Unauthorized access attempts.

Benefits of monitoring include:

- Early detection of cyber threats.
- Quick response to security incidents.
- Prevention of data breaches.
- Continuous security assessment.

Monitoring allows financial organizations to take immediate action before sensitive information is compromised.

Role of Encryption in Protecting Data

Encryption protects sensitive information by converting readable data into an unreadable format that can only be accessed using a decryption key. Cloud encryption is applied in two ways:

1. Encryption at Rest

- Protects stored data in cloud databases and storage systems.
- Prevents unauthorized access to stored financial records.

2. Encryption in Transit

- Protects data while it moves between users, applications, and cloud servers.
- Prevents attackers from intercepting transaction information.

Benefits of Encryption

- Protects customer financial data.
- Maintains confidentiality.
- Prevents unauthorized data access.
- Supports regulatory compliance.

Ensuring Data Sovereignty Requirements

- Selecting cloud regions that comply with local regulations.
- Storing customer data within approved geographical locations.
- Using cloud providers that offer regional data centers.
- Applying policies to control where data is stored and processed.

- Maintaining records of data location for auditing purposes.

Example: Sensitive Financial Transactions

Consider a banking application that processes online money transfers and customer payments. The bank stores customer account details and transaction records in the cloud. If a customer transfers money through the banking app, the transaction information is securely encrypted, monitored, and recorded. Auditing tools can verify that the transaction followed security and compliance requirements. To maintain security:

- Encryption protects transaction data during storage and transfer.
- Access control ensures only authorized banking employees can access customer information.
- Logging records every transaction and user activity.
- Monitoring systems detect unusual activities such as suspicious login attempts or abnormal transactions.

Conclusion

Cloud providers enable financial institutions to use cloud services securely while meeting strict compliance requirements. Regulatory support, auditing, logging, monitoring, and encryption protect sensitive financial information from unauthorized access and cyber threats. By selecting appropriate cloud regions and following data sovereignty policies, organizations can ensure that their data remains compliant with local laws. These security measures allow financial institutions to perform sensitive financial transactions safely and reliably in the cloud.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Introduction

Containerization is a technology that allows applications and their dependencies to be packaged together into lightweight units called containers. It helps organizations deploy applications faster, improve resource utilization, and run applications consistently across different environments. Containers are widely used in cloud computing along with orchestration tools like Kubernetes.

Containers vs Virtual Machines

Containers and virtual machines both provide application isolation, but their architecture and resource usage are different.

Virtual Machines (VMs):

- A VM runs on a physical server using a hypervisor.
- Each VM contains a complete operating system, virtual hardware, applications, and libraries.
- Requires more CPU, memory, and storage.
- Takes more time to start because the entire operating system needs to boot.
- Provides strong isolation between applications.

Containers:

- Containers share the host operating system kernel.
- Contains only application code, libraries, and required dependencies.
- Requires fewer resources compared to VMs.
- Starts quickly because there is no separate operating system.
- Allows multiple applications to run efficiently on the same server.

Performance and Resource Usage Advantages of Containers

Containers provide better performance because they have less overhead.

Advantages include:

- Faster application startup.
- Lower memory and CPU consumption.
- Better utilization of cloud resources.
- Ability to run more applications on the same hardware.
- Faster deployment and updates.

Role of Kubernetes in Container Orchestration

Managing many containers manually is difficult. Kubernetes is a container orchestration platform that automates container management. Kubernetes simplifies deployment and scaling through:

- **Automated Deployment:** Automatically deploys containers across available servers.
- **Auto-Scaling:** Increases or decreases the number of containers based on application demand.
- **Load Balancing:** Distributes user requests among multiple containers to improve performance.
- **Self-Healing:** Automatically restarts failed containers and replaces unhealthy instances.
- **Service Management:** Manages communication between different application components.

Portability Benefits Across Cloud Environments

Containers provide portability because they contain all necessary application components.

Benefits include:

- Applications can run consistently on different platforms.
- Supports public, private, and hybrid cloud environments.
- Reduces dependency on a single cloud provider.
- Makes migration between cloud platforms easier.
- Supports multi-cloud deployment strategies.

For example, a Docker container can run on AWS, Microsoft Azure, Google Cloud, or an organization's own data center without major modifications.

Example of a Microservices-Based Application

An online shopping application is an example of a microservices-based application. Each service runs independently in its own container. During high traffic periods, Kubernetes can increase the number of containers for important services such as payment and order processing. The application can be divided into separate services:

- **User Service:** Handles customer registration and profiles.
- **Product Service:** Manages product details and inventory.
- **Order Service:** Handles customer orders.
- **Payment Service:** Processes online payments.
- **Notification Service:** Sends messages and updates to customers.

Conclusion

Containerization provides a lightweight and efficient method for application deployment. Compared to virtual machines, containers use fewer resources, start faster, and provide better performance. Kubernetes simplifies container deployment, scaling, and management. The portability of containers allows applications to run across different cloud environments easily. Microservices-based applications benefit from containers by allowing individual services to be deployed and scaled independently.